3502 PRINCIPLES OF MACHINE LEARNING

Lost in Transcription

Indonesian–Javanese conversational ASR

From a broad ensemble to a domain-adapted,
conservatively fused offline system

Final displayed leaderboard value: 0.2743

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Project case study, methodology and implementation

Abstract and final outcome

Goal: transcribe spontaneous code-switching speech while preserving colloquial words, disfluencies and reference spelling.

- A broad pretrained ensemble looked strong locally but struggled on the leaderboard and runtime.
- The pivot was to adapt a reliable Whisper Turbo anchor, then accept only supported corrections.
- The final system used two Turbo checkpoints and adapted MERaLiON.

0.2743
Final displayed value; metric label unavailable

34th / 500
Team-reported final position

30m 40s
Measured CUDA inference

Key point: The achieved result belongs to the conservative Turbo system. Large-v3 is separate local research.

Evidence: Leaderboard capture omits score-column headings; the final position was supplied by the team. Submission history and CUDA log support the system and runtime claims.

The problem: conversational speech under constraints

Recognition challenges

- Indonesian and Javanese switch within a clip.
- Informal spelling, names and acronyms.
- Fillers, repeats and interrupted words.
- Conversational pauses and overlapping styles.
- Training and hidden-test domains differ.

Engineering constraints

- Development on 16 GB Apple Silicon.
- Offline execution on competition CUDA.
- Hard inference limit: two hours.
- Some target clips exceed 30 seconds.
- Every clip processed independently.
- Valid ordered CSV required for scoring.

Key point: Semantic plausibility is insufficient: wording, coverage and reliable execution all affect the result.

Data audit and domain mismatch

Dataset / prepared split	Size
Jember session recordings	208 MP3s
Original timestamped annotations	6,679 rows
Retained merged chunks	1,471
Jember train / holdout	1,309 / 162
Labelled competition clips	372
Conversation 5 / conversation 2	294 / 78

Jember: approximately **9.97 hours**.
Competition labelled audio: approximately **2.54 hours**.

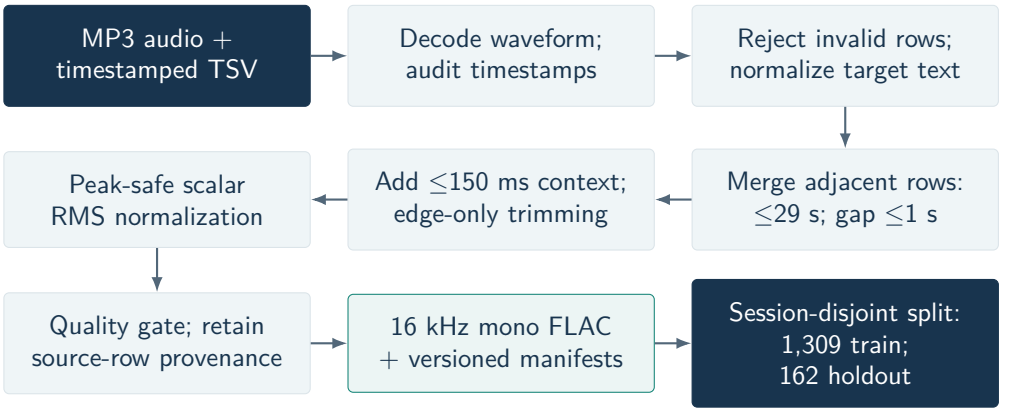
Why direct transfer was hard

- Raw Jember median: 5 s; target: 26.01 s.
- Jember is Javanese-heavy; target is mostly mixed speech.
- 352 mixed, 20 Indonesian-only, no pure-Javanese target clips.
- Prepared word-vocabulary coverage: 81.54%; this is not a subword WER floor.

Key point: Adapt duration and orthography while retaining the pretrained model's broader knowledge.

Evidence: processed/v2 manifest, duration and vocabulary audits.

Preprocessing pipeline: auditable training examples



Stage accounting: 6,679 rows → 6,638 valid → 1,472 merged → 1,471 retained.

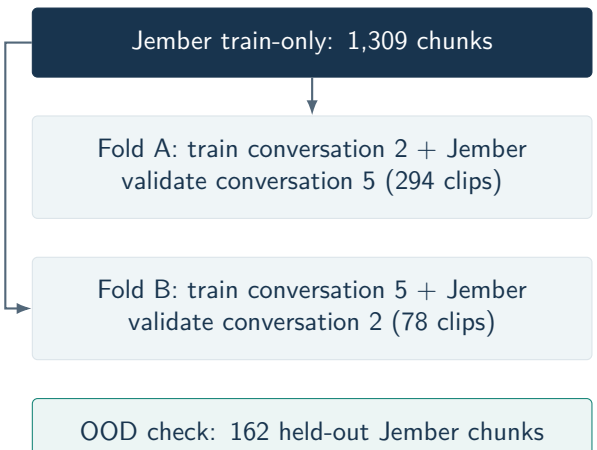
Evidence: Versioned v2 preparation; 41 invalid rows and one final chunk removed.

Preprocessing choices: preserve speech, match the task

Stage	Implemented choice	Reason / boundary
Text targets	Standardize punctuation and whitespace; remove Jember diacritics conservatively	Preserve casing, acronyms, fillers, repeats and word boundaries
Segmentation	Merge adjacent rows; pad boundaries; trim only edges	Match target duration without removing interior pauses
Audio scaling	RMS target -23 dBFS; peak cap -1 dBFS	Scalar gain, not denoising or a claim of better SNR
Quality audit	Check duration, silence and degeneracy; retain soft flags	Reject unusable examples without silently filtering hard speech
Inference audio	Decode float32; mono; resample to 16 kHz	Submitted Turbo/MER pipeline did not inherit training edge trimming or RMS scaling

Key point: Training targets, alignment keys and official scoring are separate text paths. Normalized alignment keys are not emitted as rewritten transcripts.

Evaluation: hold out entire conversations



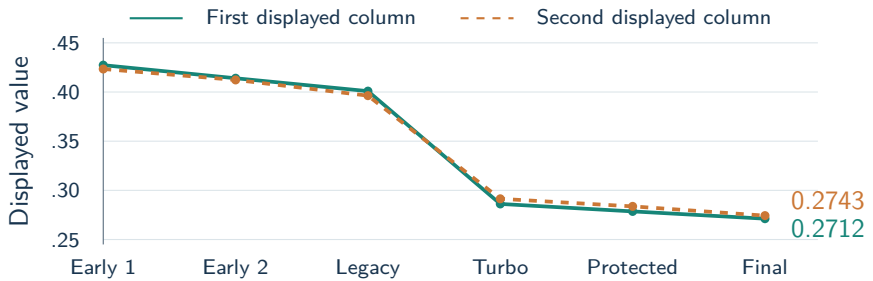
Official-compatible metric

$$\mathrm{WER} = \frac{S + D + I}{N}$$

- Aggregate = total errors / total reference words, not a fold average.
- Report both folds and worst fold.
- A clip oracle uses references to choose the best whole hypothesis; it is a diagnostic.
- Conversation identity never routes inference.

Key point: OOF reduces direct leakage, but repeated selection on only two conversations can still overfit.

Leaderboard progression: displayed values by submission stage



Second displayed column: 0.4234 at the first stage and 0.2743 at the final stage.

Key point: 0.2712 and 0.2743 belong to the same submission row. They are not two successive models.

Evidence: The archived screenshot omits both score-column headings, so these values are not identified here as WER. Final rank 34/500 is team-reported.

Initial exploration: many models, uneven usefulness

Model family	Role explored	Observed limitation
Whisper Turbo / Large	Multilingual baseline; Indonesian/Javanese conditioning	Zero-shot wording and long-form handling needed attention
Qwen3-ASR-1.7B	Generalist candidate and ensemble hypothesis	MPS throughput/cache completion problems
MERaLiON	Conversational multilingual alternative	Useful errors, but naive voting was unreliable
Javanese XLS-R	CTC specialist for Javanese speech	Dtype and decoder API failures in early deployment
Buzz Javanese	Adaptive specialist and extra voter	Weak overall; useful on a limited subset
Medoid / ROVER	Select a transcript or vote aligned words	Local gains did not establish hidden robustness

Key point: These were evolving experiments, not one fixed six-model system deployed throughout the project.

Why the broad ensemble struggled

Measured or audited evidence

- Metadata-blind local ROVER: 0.205776; legacy leaderboard: 0.4009.
- Metadata-aware replay: 0.206177. Routing alone did not explain the gap.
- ROVER was worse than the best candidate on 153 clips; better on 98.
- Early runs failed or exceeded two hours.
- Audit found stale archive/config, long-form truncation and manifest-order risks.

Plausible accuracy mechanisms

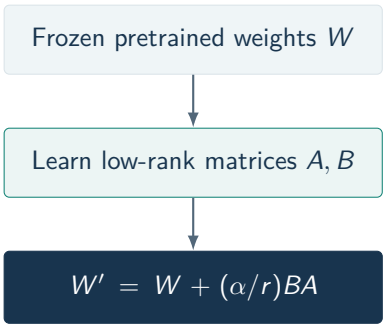
- Similar models can share wrong spelling or language preferences.
- A weak voter can damage a stronger transcript.
- Two conversations poorly represent the hidden domain.
- Repeated local tuning can specialize the selector.
- More model passes add runtime and deployment failure points.

Key point: No single cause of the hidden gap was proven. The next priority was a better anchor, not more votes.

Evidence: Legacy replay, deployment audit and preserved platform observations.

The pivot: adapt Whisper Turbo with LoRA

Component	Implemented recipe
Base	Whisper large-v3-turbo
LoRA	rank 32, alpha 64, dropout 0.05
Targets	Encoder + decoder attention / MLP
Trainable	27.85M / 836.73M (3.33%)
Optimizer	AdamW, LR 3×10^{-4}
Local compute	MPS FP32; microbatch 1
Memory strategy	Accumulate 16; checkpoint gradients
Data sampling	Competition ratio 3 + Jember
Checkpointing	Half and full weighted epoch



Retain broad acoustic knowledge while learning target-domain wording and switching patterns.

Key point: Sampling ratio changes draw weights, not the number of independent labelled recordings.

Turbo experiments: inspect both folds

Condition / ratio	Checkpoint	A	B	Aggregate	Jember
Indonesian / 3	Half	0.2236	0.2042	0.2197	0.2750
Indonesian / 3	Full	0.2259	0.1890	0.2184	0.2456
Javanese / 3	Full	0.2709	0.1842	0.2533	0.2442
Automatic / 3	Full	0.2328	0.1873	0.2235	0.2450
Indonesian / 4	Full	0.2249	0.1808	0.2159	0.2600

- Initial deployment used Indonesian ratio-3 half, prioritizing the worst fold.
- Javanese full helped B but substantially damaged A.
- Full ratio-3 later became the fusion anchor after repetition handling.
- Turbo-only leaderboard: **0.2862 / 0.2914**, the largest visible improvement step.

Key point: A lower aggregate alone did not justify a less balanced language or sampling recipe.

Evidence: Conversation-disjoint cached predictions; Jember column is the fold-model mean.

MERaLiON: adapt for complementary errors

Memory-efficient adaptation

- Decoder-only LoRA: r16 / alpha32.
- Frozen speech encoder and projector; cached representations.
- Answer-only loss; chunked vocabulary projection.
- AdamW 10^{-4}; microbatch 1, accumulation 16.
- Nonfinite full attempt retained a usable half checkpoint; stable retry used later.

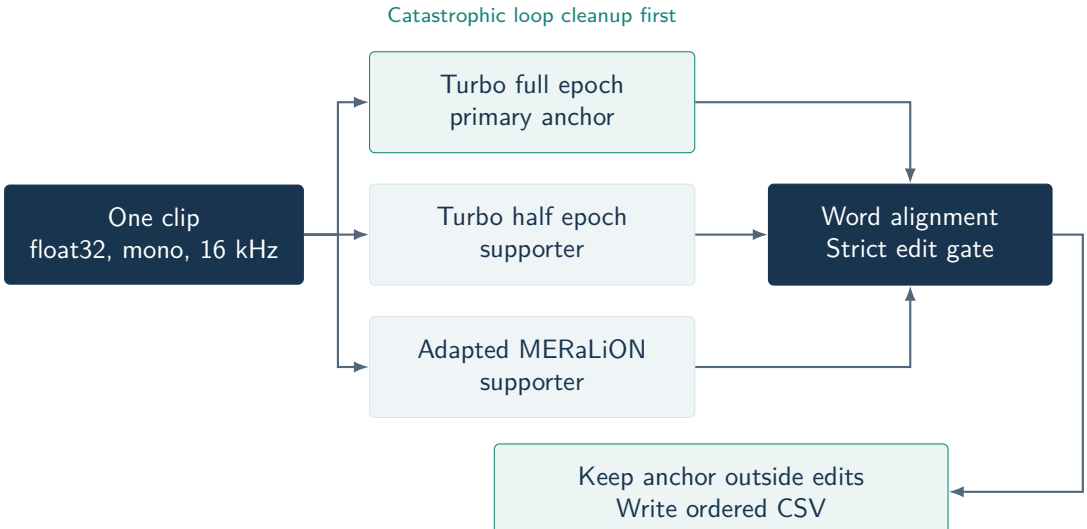
OOF system	A	B	Agg.
Whisper half	.2236	.2042	.2197
MER zero-shot	.2875	.2292	.2756
MER adapted	.2266	.2208	.2254
Adapted pair oracle	**.1892**	**.1718**	**.1857**
Simple ROVER	.2185	.2151	.2178

Complementarity is not a fusion result

The oracle improved both folds. Simple voting still made Fold B worse than Whisper alone.

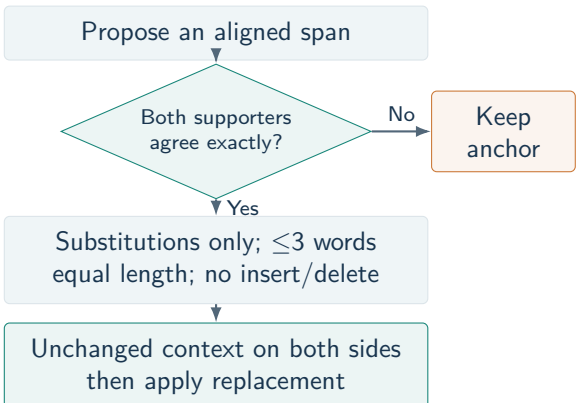
Key point: MERaLiON became a supporter for justified edits, rather than an unrestricted replacement.

Final architecture: a protected Turbo anchor



Key point: The three hypotheses are produced sequentially in deployment to control model residency.

Fusion gate: when is an edit allowed?



Safeguards

- One weak model cannot override the anchor alone.
- Reject unstable or repetitive MER support.
- Use normalized keys for alignment; preserve original surface text elsewhere.
- Single-word loops: five or more identical words collapse to two.
- If cleanup changes indices, return the cleaned anchor without further edits.

Key point: Ordinary fusion has no insertions or deletions; repetition cleanup is a separate operation.

Final selection: safer edits over a tiny OOF gain

Candidate	A	B	Aggregate	S / I / D
Full Turbo, raw	.225860	.188976	.218359	–
Full Turbo, repetition-safe	.213296	.188976	.208350	–
A: broader operations	.208342	.185321	.203660	157 / 21 / 17
B: strict substitutions	**.209778**	**.185321**	**.204804**	**134 / 0 / 0**
C: confidence gate	.210281	.187289	.205605	97 / 13 / 12

Why choose B?

61 fewer word edits than A; no ordinary insertions/deletions. Cost: only 0.001144 aggregate WER, with the same B result.

OOD sanity check

Jember diagnostic: **0.244356** for B versus 0.245552 for full Turbo. No observed collapse.

Key point: 134 substitutions in 129 spans. Shared loop cleanup removed 196 words separately.

Evidence: Official-compatible scorer. Jember uses fold-checkpoint means with final MER predictions; an OOD diagnostic, not OOF.

Error analysis: useful agreement can still be wrong

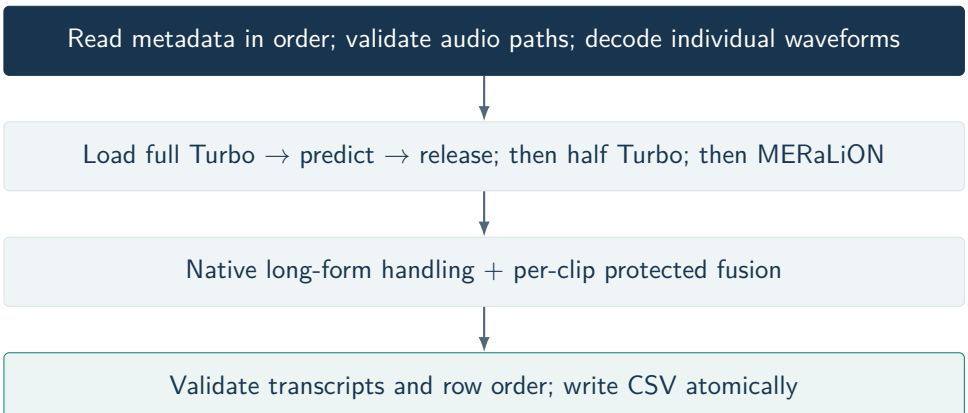
Example	Reference excerpt	Full Turbo	Fused excerpt
Helpful spelling	beberapa kali pengen	beberapa kali pengin,	beberapa kali pengen,
Helpful word	udah kayak apa sih	udah layak, apa sih,	udah kayak apa sih,
Harmful variant	Nah, nggak seperti	Nah, nggak seperti	Nah, gak seperti
Harmful override	gitu loh tau gak tau	gitu lho. Tau gak tau	gitu lho. Aku gak tau

- A supported correction can recover a reference spelling or wrong word.
- Both supporters can agree on a plausible but incorrect alternative.
- Reference-style orthography matters: `gak` and `nggak` can score differently.
- Restricting edits reduces exposure; it does not guarantee correctness.

Key point: These are labelled OOF examples used for retrospective analysis, never hidden-test inference features.

Evidence: Four saved OOF examples in docs/case_study/error_examples.json.

Deployment pipeline: offline and clip-independent



Input: `/code_execution/data/test_metadata.csv` and `data/clips/`
Output: `/code_execution/submission/submission.csv`
Schema: `audio_filename,transcript` **CUDA batch:** 8 **MPS batch:** 1

Key point: All assets are bundled locally. Batching improves throughput without cross-test adaptation or transcript logging.

Verification: a model must finish and produce valid output

Checked on the packaged system

- Model loading with sockets blocked and empty external cache.
- Normal clip: 21.60 seconds.
- Long clip: 37.65 seconds; tail retained.
- Repeated inference under fixed settings.
- Exact CSV schema and sample order.
- No blank or NaN failure rows.
- ZIP integrity and SHA256 recorded.

1,840.1 s

Measured competition CUDA run

Exit code 0

submission.csv found by runner

Archive: approximately **13.12 GB compressed**, 52 entries.

Log inventory matches the local conservative package; the upload hash is absent from that log.

Key point: Local smoke tests support correctness; the supplied CUDA completion log establishes successful platform execution.

Evidence: Latest user-supplied log, archive inventory and preserved offline smoke checks.

Large-v3 follow-up: stronger local research, separate result

Candidate	A	B	Aggregate
Submitted conservative Turbo	.209778	.185321	.204804
Large-v3 full, standalone	.205542	.178009	.199943
Large-v3 supported consensus	**.199368**	**.172666**	**.193938**

What worked locally

r16 / alpha32, BF16 frozen base + FP32 LoRA, ratio 4. Half/full delta soups tested. Full retained for B/Jember balance; Turbo + MER supplied support.

Generalization caution

The phrase-loop guard changed Jember standalone WER from 0.265660 to 0.246524 and fused WER from 0.261175 to 0.242039; it changed neither OOF fold. Beam 3 and learned confidence gates were not selected.

Key point: No supplied leaderboard result is attributed to this package. It did not produce the reported 0.2743.

Evidence: Overnight and final nine-hour reports; preserved fold caches and checkpoint selection results.

Large-v3 ablations and offline package status

Adapter contribution sweep

Coefficient	Stand. OOF	Worst fold	Jember
0.00	0.211438	0.216886	0.268351
0.25	0.205891	0.211286	0.259755
0.50	0.200400	0.205471	0.253476
0.75	**0.199657**	0.204824	0.252429
1.00	0.199943	**0.205542**	**0.250635**

Why retain the full adapter?

The 0.75 soup had a one-error aggregate edge; the full adapter was better on Fold B and Jember and used a simpler delta.

N-best rescoring: tested, not selected

OOF aggregate moved from 0.193938 to 0.193423, but the gain was below the pre-set 0.0015 threshold. It made 305 substitutions, 2 insertions and 8 deletions; consensus made 277 substitutions and no insertions/deletions.

Package status

Offline smoke and archive checks passed, including a 37.65-second clip. A100 runtime is estimated at 60–100 minutes, not measured. This Large-v3 package was not submitted.

Key point: The selected Large-v3 result remains local OOF evidence; it has no leaderboard score.

Evidence: FINAL_9H_REPORT.md and final_9h_attack result logs.

Limitations and lessons

Limits of the evidence

- Only two target conversations.
- Repeated selection on the same folds.
- No pure-Javanese target benchmark.
- Jember is not the hidden distribution.
- Agreement is not independent truth.
- No established statistical significance.
- Turbo ratio-4 stopped at batch 218/2,797; no evaluation was completed.

What changed the approach

- Improve the anchor before expanding fusion.
- Audit data and deployment together.
- Check both folds and edit aggressiveness.
- Use oracles to diagnose, not claim deployable performance.
- Preserve caches, checkpoints and hashes.
- Keep recovery and runtime gates explicit.

Key point: The local-to-hidden gap remains unresolved; the measured improvement does not prove every design choice caused it. The stopped Turbo run has a saved resume state.

Evidence: The ratio-4 checkpoint and stop status are documented in phase4_stronger_anchor/RESUME.md.

Conclusion and evidence trail

Delivered solution

- Versioned corpus preparation and official-compatible evaluation.
- Domain-adapted Turbo full anchor with half-Turbo and MER support.
- Conservative substitution gate and repetition protection.
- A successful offline CUDA competition package.

Next research step: independent conversation validation for Large-v3 and its error corrections before claiming hidden robustness.

34th / 500

Team-reported final leaderboard position

0.204804

Submitted system local OOF WER

AI tools assisted research, coding and documentation; claims are grounded in saved experiment artifacts.

Evidence / references: Project case study; processed/v2 manifest; fold caches; conservative selection report; latest CUDA log and submission screenshot.

Hu et al., LoRA (2021) · OpenAI Whisper Turbo model card · DrivenData competition runtime